

Anser

Ускоряем запросы

Идея: в распределённом плане запроса с Motion и Hash Join заранее отфильтруем большую таблицу значениями из меньшей.

Реализация: автоматически создадим Bloom-фильтры на основе результатов сканирования меньшей таблицы и применим их при сканировании большей.



Оригинальная статья



Cloudberry PR #1942

```
postgres=# explain analyze select aef.name ,aef.value ,aef.id ,a.* from applications_extra_fields as aef left join applications as a on aef.id = a.id;
```

QUERY PLAN

```
Gather Motion 3:1 (slice1; segments: 3) (cost=0.00..1319.53 rows=20000 width=385) (actual time=45.030..244.790 rows=10000 loops=1)
-> Hash Right Join (cost=0.00..1290.83 rows=6667 width=385) (actual time=46.043..233.784 rows=3400 loops=1)
    Hash Cond: (a.id = aef.id)
    Extra Text: (seq2) Hash chain length 10.0 avg. 10 max. using 340 of 262144 buckets.
    -> Custom Scan (Anser Bloom Consumer) (cost=0.00..499.02 rows=333334 width=353) (actual time=33.550..218.237 rows=340 loops=1)
        Bloom Filter Size: 1048576 bytes
        Bloom Filter Stats: memory=1024kB checked=334042 rejected=333702
        Rows Removed by Bloom Filter: 333702
        -> Seq Scan on applications a (cost=0.00..499.02 rows=333334 width=353) (actual time=1.921..150.350 rows=334042 loops=1)
    -> Hash (cost=431.62..431.62 rows=3334 width=32) (actual time=10.895..10.896 rows=3400 loops=1)
        Buckets: 262144 Batches: 1 Memory Usage: 2265kB
        -> Redistribute Motion 3:3 (slice2; segments: 3) (cost=0.00..431.62 rows=3334 width=32) (actual time=0.768..10.359 rows=3400 loops=1)
            Hash Key: aef.id
            -> Custom Scan (Anser Bloom Producer) (cost=0.00..431.09 rows=3334 width=32) (actual time=0.591..9.212 rows=5000 loops=1)
                Bloom Filter Size: 1048576 bytes
                Bloom Filter Stats: memory=1024kB
                -> Seq Scan on applications_extra_fields aef (cost=0.00..431.09 rows=3334 width=32) (actual time=0.589..1.247 rows=5000 loops=1)
        )
Planning Time: 16.477 ms
(slice0) Executor memory: 1143K bytes.
(slice1) Executor memory: 6544K bytes avg x 3x(0) workers, 6555K bytes max (seg0). Work_mem: 2265K bytes max.
(slice2) Executor memory: 6467K bytes avg x 3x(0) workers, 6467K bytes max (seg2).
Memory used: 128000kB
Optimizer: GPORCA
Execution Time: 252.013 ms
(24 rows)
```